Chapter 2. Permutations, Cosets, and Direct Products

Permutation groups, Cayley, cycles and the alternating group, Lagrange, direct products, finitely generated abelian groups, plane isometries

Contents

2.1 Groups of permutations (§8) **2**

2.2 Orbits, cycles, and the alternating groups (§9) **2**

2.3 Cosets and the theorem of Lagrange (§10) **3**

2.4 Direct products and finitely generated abelian groups (§11) **4**

2.5 Plane isometries (§12) **6**

2.1 Groups of permutations (§8)

A **permutation** of a set A is a bijection $A \to A$; their product $\sigma\tau$ is the composition $\sigma \circ \tau$ (apply τ first).

Theorem 2.1 (8.5)

The permutations of a nonempty set A form a group S_A under composition.

Proof. A composite of bijections is a bijection, composition is associative (Theorem 1.2), the identity map is an identity, and the inverse function of a bijection is an inverse. ■

$S_n = S_{\{1,\ldots,n\}}$ is the **symmetric group**, of order $n!$; it is nonabelian for $n \geq 3$. The **dihedral group** D_n ($n \geq 3$) is the group of symmetries of a regular n-gon, a subgroup of S_n of order $2n$; $D_3 = S_3$.

Lemma 2.2 (8.15)

If $\phi : G \to G'$ is one-to-one and $\phi(xy) = \phi(x)\phi(y)$ for all x, y, then $\phi[G]$ is a subgroup of G' and ϕ is an isomorphism of G with $\phi[G]$.

Proof. $\phi(x)\phi(y) = \phi(xy) \in \phi[G]$. From $\phi(e)\phi(e) = \phi(e) = e'\phi(e)$, cancellation gives $\phi(e) = e'$; then $\phi(a)\phi(a^{-1}) = \phi(e) = e'$ gives $\phi(a)^{-1} = \phi(a^{-1}) \in \phi[G]$. So $\phi[G] \leq G'$ (Theorem 1.8), and $\phi : G \to \phi[G]$ is a bijection preserving the operation. ■

Theorem 2.3 (8.16, Cayley's theorem)

Every group is isomorphic to a group of permutations.

Proof. For $x \in G$ let $\lambda_x : G \to G$, $\lambda_x(g) = xg$. It is one-to-one ($xg_1 = xg_2 \Rightarrow g_1 = g_2$) and onto ($\lambda_x(x^{-1}c) = c$), so $\lambda_x \in S_G$. The map $\phi(x) = \lambda_x$ is one-to-one ($\lambda_x(e) = x$) and $\lambda_{xy}(g) = xyg = \lambda_x(\lambda_y(g))$, so $\phi(xy) = \phi(x)\phi(y)$. Apply Lemma 2.2. ■

ϕ is the **left regular representation**; with $\rho_x(g) = gx^{-1}$ one gets the right regular representation.

2.2 Orbits, cycles, and the alternating groups (§9)

For $\sigma \in S_A$, $a \sim b$ iff $b = \sigma^n(a)$ for some $n \in \mathbb{Z}$ is an equivalence relation; its cells are the **orbits** of σ. A **cycle** is a permutation with at most one orbit of more than one element; $(a_1, a_2, \ldots, a_m)$ sends $a_i \mapsto a_{i+1}$, $a_m \mapsto a_1$, its **length** is m; a cycle of length 2 is a **transposition**.

Theorem 2.4 (9.8)

Every permutation of a finite set is a product of disjoint cycles.

Proof. Let $B_1, \ldots, B_r$ be the orbits of σ and μ_i the cycle that agrees with σ on B_i and is the identity elsewhere (on an orbit $B = \{b, \sigma(b), \ldots, \sigma^{m-1}(b)\}$, σ acts as the cycle $(b, \sigma(b), \ldots, \sigma^{m-1}(b))$). The orbits are disjoint, so $\sigma = \mu_1\mu_2 \cdots \mu_r$: each x lies in exactly one B_i, and only μ_i moves it. ■

Disjoint cycles commute, and the decomposition is unique up to order (the cycles are determined by the orbits).

Corollary 2.5 (9.12)

Any permutation of a finite set with at least two elements is a product of transpositions.

Proof. $(a_1, a_2, \ldots, a_m) = (a_1, a_m)(a_1, a_{m-1}) \cdots (a_1, a_2)$, and the identity is $(1, 2)(1, 2)$. ■

Theorem 2.6 (9.15)

No permutation in S_n is both a product of an even number and of an odd number of transpositions.

Proof. (Orbit counting.) Let $\sigma \in S_n$ have r orbits (counting fixed points) and set $N(\sigma) = n - r$. For a transposition $\tau = (i, j)$, $N(\tau\sigma) = N(\sigma) \pm 1$: if i, j lie in the same orbit of σ, write the cycle as $(i, b_1, \ldots, b_h, j, c_1, \ldots, c_k)$; then (i, j) times it is $(i, c_1, \ldots, c_k)(j, b_1, \ldots, b_h)$, so the number of orbits increases by 1. If i, j lie in different orbits, the same computation read backwards shows the two orbits merge, decreasing the count by 1. Since $N(\mathrm{id}) = 0$, a product of k transpositions has $N \equiv k \pmod 2$. So the parity of k is determined by σ. ■

A permutation is **even** or **odd** according to this parity; a cycle of length m has parity of $m - 1$.

Theorem 2.7 (9.20)

For $n \geq 2$, the even permutations of $\{1, \ldots, n\}$ form a subgroup A_n of S_n of order $n!/2$.

Proof. Products of even permutations are even, the identity is even, and the inverse of $\tau_1 \cdots \tau_k$ is $\tau_k \cdots \tau_1$; so $A_n \leq S_n$. Fix $\tau = (1, 2)$. $\lambda_\tau(\sigma) = \tau\sigma$ maps A_n into the odd permutations B_n and B_n into A_n, and $\lambda_\tau^2 = \mathrm{id}$; so it is a bijection $A_n \to B_n$ and $|A_n| = |B_n| = n!/2$. ■

A_n is the **alternating group**.

2.3 Cosets and the theorem of Lagrange (§10)

Theorem 2.8 (10.1)

Let $H \leq G$. The relations $a \sim_L b \iff a^{-1}b \in H$ and $a \sim_R b \iff ab^{-1} \in H$ are equivalence relations on G.

Proof. For $\sim_L$: $a^{-1}a = e \in H$; if $a^{-1}b \in H$ then $(a^{-1}b)^{-1} = b^{-1}a \in H$; if $a^{-1}b, b^{-1}c \in H$ then their product $a^{-1}c \in H$. $\sim_R$ is analogous. ■

The cell of a under $\sim_L$ is the **left coset** $aH = \{ah : h \in H\}$ (as $a^{-1}b = h \iff b = ah$); under $\sim_R$ it is the **right coset** Ha. For abelian G they coincide.

Theorem 2.9 (10.10, Lagrange)

If H is a subgroup of a finite group G, then $|H|$ divides $|G|$.

Proof. $h \mapsto ah$ is a bijection $H \to aH$ (one-to-one by cancellation), so every left coset has $|H|$ elements. The left cosets partition G (Theorems 2.8, 1.1); if there are r of them, $|G| = r|H|$. ■

Corollary 2.10 (10.11)

Every group of prime order is cyclic.

Proof. For $a \neq e$, $|\langle a\rangle| > 1$ divides p, so equals p and $\langle a\rangle = G$. ■

Theorem 2.11 (10.12)

The order of an element of a finite group divides the order of the group.

Proof. It is the order of the cyclic subgroup it generates. ■

The number of left cosets of H in G is the **index** $(G : H)$; the map $aH \mapsto Ha^{-1}$ is a bijection between left and right cosets, so the index is the same for both.

Theorem 2.12 (10.14)

If $K \leq H \leq G$ with $(G : H)$ and $(H : K)$ finite, then $(G : K)$ is finite and $(G : K) = (G : H)(H : K)$.

Proof. Let $G = \bigsqcup_{i\leq r} a_iH$ and $H = \bigsqcup_{j\leq s} b_jK$. Then $a_iH = \bigsqcup_j a_ib_jK$, so $G = \bigcup_{i,j} a_ib_jK$. These rs cosets are distinct: if $a_ib_jK = a_pb_qK$, then since $b_jK, b_qK \subset H$, the cosets a_iH and a_pH meet, so $i = p$; then $b_jK = b_qK$ and $j = q$. ■

The converse of Lagrange fails: A_4 (order 12) has no subgroup of order 6.

2.4 Direct products and finitely generated abelian groups (§11)

Theorem 2.13 (11.2)

For groups $G_1, \ldots, G_n$, $\prod G_i$ is a group (the **direct product**) under $(a_1, \ldots, a_n)(b_1, \ldots, b_n) = (a_1b_1, \ldots, a_nb_n)$.

Proof. The operation is componentwise, so associativity holds componentwise; $(e_1, \ldots, e_n)$ is the identity and $(a_1^{-1}, \ldots, a_n^{-1})$ the inverse. ■

Theorem 2.14 (11.5)

$\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic, and isomorphic to $\mathbb{Z}_{mn}$, iff $\gcd(m, n) = 1$.

Proof. If $\gcd(m, n) = 1$, the order of $(1, 1)$ is the least k divisible by both m and n, namely mn, so $(1, 1)$ generates. If $d = \gcd(m, n) > 1$, then mn/d is divisible by m and n, so every element (r, s) satisfies $(mn/d)(r, s) = (0, 0)$; no element has order mn. ■

Corollary 2.15 (11.6)

$\prod_{i=1}^n \mathbb{Z}_{m_i}$ is cyclic and isomorphic to $\mathbb{Z}_{m_1\cdots m_n}$ iff the m_i are pairwise relatively prime.

Proof. Induction using Theorem 2.14 and $\prod_{i\leq n} \mathbb{Z}_{m_i} \cong (\prod_{i<n} \mathbb{Z}_{m_i}) \times \mathbb{Z}_{m_n}$: if pairwise relatively prime, $m_1 \cdots m_{n-1}$ and m_n are relatively prime. Conversely if $\gcd(m_i, m_j) > 1$, the subgroup $\mathbb{Z}_{m_i} \times \mathbb{Z}_{m_j}$ is not cyclic, but subgroups of cyclic groups are cyclic. ■

Theorem 2.16 (11.9)

If $a_i \in G_i$ has finite order r_i, then $(a_1, \ldots, a_n)$ has order $\operatorname{lcm}(r_1, \ldots, r_n)$.

Proof. $(a_1,\dots,a_n)^k = (e_1,\dots,e_n)$ iff $r_i \mid k$ for all i; the least such $k>0$ is the lcm. ■

Theorem 2.17 (11.12, fundamental theorem of finitely generated abelian groups)

Every finitely generated abelian group is isomorphic to

$$\mathbb{Z}_{(p_1)^{r_1}} \times \mathbb{Z}_{(p_2)^{r_2}} \times \cdots \times \mathbb{Z}_{(p_n)^{r_n}} \times \mathbb{Z} \times \cdots \times \mathbb{Z},$$

with primes p_i (not necessarily distinct) and $r_i \in \mathbb{Z}^+$. The direct product is unique except for the order of the factors; the number of factors $\mathbb{Z}$ is the **Betti number**.

Proof.

Step 1. (Stacked bases.) Write G additively, generated by $x_1,\dots,x_n$. The map $\mathbb{Z}^n \to G$, $(m_i) \mapsto \sum m_i x_i$, is an onto homomorphism with kernel K, and $G \cong \mathbb{Z}^n/K$ (Theorem 3.9). We show: there are a basis $f_1,\dots,f_n$ of $\mathbb{Z}^n$ and integers $d_1 \mid d_2 \mid \cdots \mid d_k$ ($k \le n$, $d_i > 0$) such that $d_1 f_1,\dots,d_k f_k$ is a basis of K. Induction on n; if $K=0$ take $k=0$. Over all bases $f_1,\dots,f_n$ of $\mathbb{Z}^n$ and all nonzero $v = \sum c_i f_i \in K$, choose one in which some coefficient has the least possible positive absolute value d; renumbering and changing signs, $c_1 = d$.

(i) d divides every c_j: if $c_j = qd + r$ with $0 < r < d$, the basis $f_1 + qf_j, f_2, \dots, f_n$ (still a basis, as $f_1 = (f_1 + qf_j) - qf_j$) gives v the j-th coefficient $c_j - qd = r$, contradicting minimality. So $v = dg_1$ with $g_1 = f_1 + \sum_{i>1}(c_i/d)f_i$, and $g_1, f_2, \dots, f_n$ is a basis.

(ii) For $w = ag_1 + \sum_{i>1} b_i f_i \in K$, d divides a (else $w - qv$ has coefficient $a - qd \in (0,d)$), and d divides every b_i (apply (i) to the element $v + (w - \frac{a}{d}v) = dg_1 + \sum b_i f_i$ of K). Hence $K = \mathbb{Z}\, dg_1 \oplus K'$ with $K' = K \cap \langle f_2,\dots,f_n\rangle$, and d divides all coefficients of elements of K'.

(iii) By induction, $\langle f_2,\dots,f_n\rangle \cong \mathbb{Z}^{n-1}$ has a basis $g_2,\dots,g_n$ with $d_2 g_2,\dots,d_k g_k$ a basis of K' and $d_2 \mid \cdots \mid d_k$. The coordinates of g_2 with respect to $f_2,\dots,f_n$ have gcd 1 (it is part of a basis), and d divides all coordinates of $d_2 g_2$, so $d \mid d_2$. Put $d_1 = d$.

Step 2. (Existence.) Then $G \cong \mathbb{Z}^n/K \cong \mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_k} \times \mathbb{Z}^{n-k}$, via $\sum m_i f_i \mapsto (m_1 \bmod d_1, \dots, m_k \bmod d_k, m_{k+1}, \dots, m_n)$, an onto homomorphism with kernel K. Drop factors with $d_i = 1$, and split each $\mathbb{Z}_d$ with $d = \prod p_j^{s_j}$ as $\prod \mathbb{Z}_{p_j^{s_j}}$ (Corollary 2.15).

Step 3. (Uniqueness.) The torsion subgroup T (elements of finite order) corresponds to the finite factors, and $G/T \cong \mathbb{Z}^b$, where b is the number of $\mathbb{Z}$ factors; b is determined since $\mathbb{Z}^b/2\mathbb{Z}^b$ has 2^b elements. For the finite part and a prime p, let T_p be the elements of p-power order, the product of the factors $\mathbb{Z}_{p^r}$ with this p. If $T_p \cong \prod_j \mathbb{Z}_{p^{r_j}}$, then $p^{k-1}T_p/p^kT_p$ has order p^{N_k}, where N_k is the number of j with $r_j \ge k$. Since these groups depend only on G, the multiset $\{r_j\}$ is determined. ■

Example 2.1

Abelian groups of order $360 = 2^3 \cdot 3^2 \cdot 5$, up to isomorphism: $3 \cdot 2 \cdot 1 = 6$ of them (partitions of 3 for the 2-part, of 2 for the 3-part), e.g. $\mathbb{Z}_8 \times \mathbb{Z}_9 \times \mathbb{Z}_5$ and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$.

A group is **decomposable** if it is isomorphic to a direct product of two proper nontrivial subgroups.

Theorem 2.18 (11.15)

The finite indecomposable abelian groups are exactly the cyclic groups of prime power order.

Proof. By Theorem 2.17 a finite indecomposable abelian group has one factor $\mathbb{Z}_{p^r}$. Conversely, if $\mathbb{Z}_{p^r} \cong H \times K$ with nontrivial H, K of orders p^s, p^t ($s,t < r$), then every element has order at most $p^{\max(s,t)} < p^r$, contradicting cyclicity. ■

Theorem 2.19 (11.16)

If m divides the order of a finite abelian group G, then G has a subgroup of order m.

Proof. Write $G \cong \prod \mathbb{Z}_{(p_i)^{r_i}}$ and $m = \prod_p p^{t_p}$ with $t_p \le \sum_{p_i=p} r_i$. For each prime p choose $s_i \le r_i$ (for $p_i = p$) with $\sum s_i = t_p$. The subgroup $\langle (p_i)^{r_i - s_i} \rangle \le \mathbb{Z}_{(p_i)^{r_i}}$ has order $(p_i)^{s_i}$, and the product of these subgroups has order m. ■

Theorem 2.20 (11.17)

If m is square free, every abelian group of order m is cyclic.

Proof. In Theorem 2.17 the primes p_i must be distinct with $r_i = 1$; then $\prod \mathbb{Z}_{p_i} \cong \mathbb{Z}_m$ by Corollary 2.15. ■

2.5 Plane isometries (§12)

An **isometry** of $\mathbb{R}^2$ is a distance-preserving map. Every isometry is a translation, a rotation, a reflection, or a glide reflection; it is affine, $\phi(x) = Ax + c$ with A orthogonal (an isometry fixing 0 preserves norms, hence inner products by polarization, hence is linear). Orientation-preserving isometries ($\det A = 1$) are translations and rotations.

Theorem 2.21 (12.5)

Every finite group G of isometries of the plane is isomorphic to $\mathbb{Z}_n$ or to a dihedral group D_n.

Proof.

Step 1. Let $G = \{\phi_1, \dots, \phi_m\}$ and $P = \frac{1}{m}\sum_i \phi_i(0)$. Since each ϕ is affine, $\phi(P) = \frac{1}{m}\sum_i \phi(\phi_i(0)) = \frac{1}{m}\sum_i \phi_i(0) = P$ ($\phi\phi_i$ runs over G). Translating, assume $P = 0$; then G consists of rotations about 0 and reflections in lines through 0.

Step 2. The rotations form a subgroup H (a product of two orientation-preserving isometries is one). If $H \neq \{\mathrm{id}\}$, let ρ be the rotation in H by the least positive angle θ. Every rotation in H, by angle $\alpha \in [0, 2\pi)$, satisfies $\alpha = k\theta + \beta$ with $0 \le \beta < \theta$, and ρ^{-k} composed with it rotates by β, so $\beta = 0$; hence $H = \langle \rho \rangle \cong \mathbb{Z}_n$.

Step 3. If $G = H$ we are done. Otherwise let $\mu \in G$ be a reflection. For any reflection $\mu' \in G$, $\mu^{-1}\mu'$ preserves orientation, so lies in H; hence $G = H \cup \mu H$ has order $2n$. Since $\mu\rho\mu^{-1} = \rho^{-1}$, G is generated by ρ (order n) and μ (order 2) with this relation, which is the dihedral group D_n (for $n = 1$, $\mathbb{Z}_2$; for $n = 2$, the Klein group D_2). ■